

Date _____
Page _____

Beta Distribution: [First kernel (B_1)]

$$\text{Beta function, } \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, \quad m > 0, n > 0$$

The relationship between beta and Gamma function is,

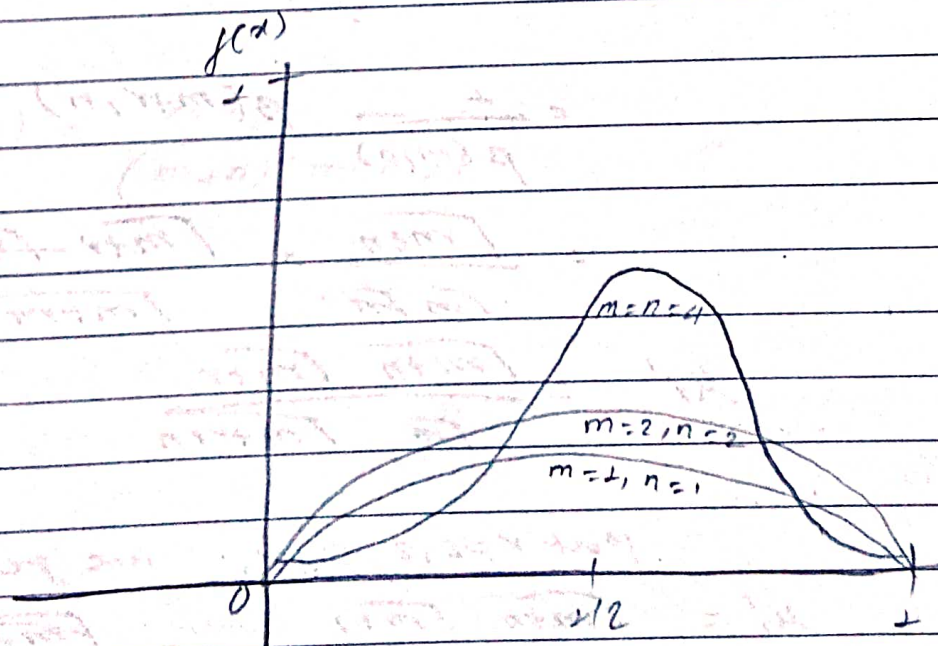
$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

A random variable x is said to have a Beta for distribution of first kernel with parameters m and n if its pdf is defined as,

$$f(x) = \frac{x^{m-1} (1-x)^{n-1}}{\beta(m, n)}, \quad 0 < x < 1$$

$m > 0$
 $n > 0$

$= 0$, otherwise



The shape of the Beta curve depends on the values of m and n when the

parameters m and n are equal, the Beta curve $y=f(x)$ is symmetrical about the

→ If $m > n$, the Beta curve is skewed to the left

→ If $m < n$, the Beta curve is skewed to the right.

Moments of Beta distⁿ of first kind:

Let $x \sim \beta_1(m, n)$, then the r th moment about origin of Beta distribution of first kind is,

$$\mu_r' = E(x^r) = \int_0^1 x^r f(x) dx$$

$$= \int_0^1 x^r \frac{x^{m-1} (1-x)^{n-1}}{\beta(m, n)} dx$$

$$= \frac{1}{\beta(m, n)} \int_0^1 x^{m+r-1} (1-x)^{n-1} dx$$

$$= \frac{1}{\beta(m, n)} \cdot \beta(m+r, n)$$

$$= \frac{\Gamma(m+n)}{\Gamma(m) \Gamma(n)} \times \frac{\Gamma(m+r) \Gamma(n)}{\Gamma(m+r+n)}$$

$$\mu_r' = \frac{\Gamma(m+n) \Gamma(m+r)}{\Gamma(m) \Gamma(m+r+n)}$$

Put $r=1, 2, 3, 4$, we get,

$$\mu_1' = \frac{\Gamma(m+n) \Gamma(m+1)}{\Gamma(m) \Gamma(m+n+1)} = \frac{\Gamma(m+n) m \Gamma(m)}{\Gamma(m) \cdot (m+n) \Gamma(m+n)}$$

$$\boxed{E(x) = \frac{m}{m+n}}$$

$$\mu_2' = \frac{\sqrt{m+n} \sqrt{m+2}}{\sqrt{m} \sqrt{m+2+n}} = \frac{\sqrt{m+n} (m)(m+2) \sqrt{n}}{\sqrt{m} (m+2+n) \sqrt{m+n}}$$

$$\mu_2' = \frac{m(m+2)}{(m+2+n)(m+n)}$$

$$v(x) = \mu_2' - (\mu_1')^2$$

$$= \frac{m(m+2)}{(m+2+n)(m+n)} - \left(\frac{m}{m+n} \right)^2$$

$$= \frac{m}{m+n} \left[\frac{m+2}{m+n+1} - \frac{m}{m+n} \right]$$

$$= \frac{m}{m+n} \left[\frac{m(m+n) - m(m+1)}{(m+n+1)(m+n)} \right]$$

$$= \frac{m}{m+n} \left[\frac{m+n-1}{(m+n+1)(m+n)} \right] \quad \begin{matrix} m^2 + mn + mn \\ - m^2 - mn - m \end{matrix}$$

$$v(x) = \frac{mn}{(m+n)^2(m+n+1)}$$

Mode:
mode is the solution of $f'(x) = 0$ and $f''(x) < 0$.

Here, $X \sim \beta(m, n)$ then its pdf is

$$f(x) = \frac{1}{\beta(m, n)} x^{m-1} (1-x)^{n-1} \quad 0 < x < 1$$

Diff. w.r. to x

$$f'(x) = \frac{1}{\beta(m, n)} \left[x^{m-1} (n-1) x^{m-2} (-1) + (1-x)^{n-1} (m-1) x^{m-1} \right]$$

$$= \frac{1}{\beta(m, n)} \left[(1-x)^{n-1} (m-1) x^{m-2} - x^{m-1} (n-1) x^{n-2} \right]$$

$$= \frac{1}{\beta(m, n)} \left[x^{m-2} (1-x)^{n-1} \left\{ (m-1) - \frac{(n-1)x}{1-x} \right\} \right]$$

mode ~~is~~ $f'(x) = 0$

$$(m-1) - \frac{(n-1)x}{1-x} = 0$$

$$\therefore x = \frac{m-1}{m+n-2} \quad \text{for } m > 1, n > 1$$

$$\therefore M_0 = \frac{m-1}{m+n-2} \quad \text{so that } f''(x) < 0$$

If $m < 1$ then $x = 0$

$n < 1$ then $x = 1$

That means Beta distribution of 1st kind is Bimodal.

Example: let $X \sim B_1(m=n=3)$ then $P(X \leq 0.5) = ?$
Soln:

let $X \sim B_1(m, n)$ (first kind) then its probability density function (pdf) is

$$f(x) = \frac{1}{B(m, n)} x^{m-1} (1-x)^{n-1}, \quad 0 < x < 1$$

$$= 0, \text{ otherwise}$$

If $X \sim B_1(3, 3)$ then its pdf,

$$f(x) = \frac{\Gamma 6}{\Gamma 3 \Gamma 3} x^{3-1} (1-x)^{3-1}$$

$$= \frac{5!}{2!2!} x^2 (1-x)^2$$

$$f(x) = 30(x^2 - 2x^3 + x^4)$$

The prob. that X is not greater than 0.5,

$$P(X \leq 0.5) = \int_0^{0.5} f(x) dx$$

$$= \int_0^{0.5} 30(x^2 - 2x^3 + x^4) dx$$

$$= 0.5$$

Example: let $X \sim B_1(3, 4)$ then find mean, mode and variance of the Beta distribution of 1st kind.

Soln:

$$X \sim B_1(m, n)$$

$$X \sim B_1(3, 4)$$

Now,

$$E(x): \text{Mean} = \frac{m+n}{m+n} = 3/7$$

$$\text{Variance} = \frac{mn}{(m+n)^2(m+n+1)} = \frac{3 \times 4}{(3+4)^2(3+4+1)} = \frac{12}{49 \times 8} = \frac{3}{98}$$

$$\text{Mode } (M_0) = \frac{m-1}{m+n-2} = \frac{2}{5}$$

Harmonic Mean:

Let $X \sim B_1(m, n)$ then the harmonic mean of the Beta distribution of first kind is given by,

$$\frac{1}{H} = E\left(\frac{1}{X}\right)$$

$$= \int_0^1 \frac{1}{x} f(x) dx$$

$$= \int_0^1 \frac{1}{x}$$

Here $X \sim B_1(m, n)$ then its pdf is

$$f(x) = \frac{1}{B(m, n)} x^{m-1} (1-x)^{n-1}$$

$$\text{Then, } \frac{1}{H} = \int_0^1 \frac{1}{x} \cdot \frac{1}{B(m, n)} x^{m-1} (1-x)^{n-1} dx$$

$$= \frac{1}{B(m, n)} \int_0^1 x^{m-2} (1-x)^{n-1} dx$$

$$= \frac{1}{B(m, n)} B(m-1, n)$$

$$= \frac{\Gamma(m+n)}{\Gamma(m) \Gamma(n)} \times \frac{\Gamma(m-1) \Gamma(n)}{\Gamma(m+n-1)}$$

$$= \frac{\sqrt{m+n}}{\sqrt{m}} \times \frac{\sqrt{m-1}}{\sqrt{m-1+n}}$$

$$= \frac{(m+n-1) \sqrt{m+n-1}}{(m-1) \sqrt{m-1}} \times \frac{\sqrt{m-1}}{\sqrt{m+n-1}}$$

$$\boxed{\frac{1}{H} = \frac{m+n-1}{m-1}}$$

$$\therefore \boxed{H = \frac{m-1}{m+n-1}}$$

for HM:

$$HM = \frac{m-1}{m+n-1}$$

$$HM = 1 - \frac{m}{m+n-1}$$

$$\boxed{HM = \frac{n}{m+n-1}}$$

For AM:

$$AM = \frac{m}{m+n}$$

$$AM = 1 - \frac{m}{m+n}$$

$$\boxed{AM = \frac{n}{m+n}}$$

That means, $HM > AM$ in $B_1(m, n)$.

Prove that: $\int_0^1 x^m (1-x)^n dx = \frac{m!n!}{(m+n+1)!}$

Proof:

we have,

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\text{Put } x = \cos^2 \theta$$

$$dx = 2 \cos \theta (-\sin \theta) d\theta$$

$$\text{when } x=0, \text{ then } \theta = \pi/2$$

$$x=1, \text{ then } \theta = 0$$

now

$$\beta(m, n) = \int_{\pi/2}^0 (\cos 2\theta)^{m-1} (1 - \cos 2\theta)^{n-1} 2 \cos \theta (-\sin \theta) d\theta$$

$$= \int_0^{\pi/2} 2 \int_0^{\pi/2} (\cos 2\theta)^{m-1} \cos \theta (\sin^2 \theta)^{n-1} \sin \theta d\theta$$

$$= 2 \int_0^{\pi/2} \cos^{2m-2} \theta \cdot \cos \theta \sin^{2n-2} \theta \sin \theta d\theta$$

$$= 2 \int_0^{\pi/2} \cos^{2m-1} \theta \sin^{2n-1} \theta d\theta$$

$$\beta(1/2, 1/2) = 2 \int_0^{\pi/2} \cos^{2 \cdot 1/2 - 1} \theta \sin^{2 \cdot 1/2 - 1} \theta d\theta$$

$$= 2 \int_0^{\pi/2} d\theta = 2 [\theta]_0^{\pi/2}$$

$$= 2 [\pi/2 - 0]$$

$$\beta(1/2, 1/2) = \pi$$

Again, $\beta(1/2, 1/2) = \frac{\Gamma(1/2) \Gamma(1/2)}{\Gamma(1/2 + 1/2)}$

$$\Rightarrow \pi = (\Gamma(1/2))^2$$

$$\Rightarrow \Gamma(1/2) = \sqrt{\pi} \text{ proved.}$$

Example: Let X be a random variable with pdf $f(x) = \frac{ge^{-3x}}{\sqrt{2}}$ or x^{2-1} , $x > 0$

- (a) Find (i) $P(X < 3)$ (ii) $P(X > 3)$
(iii) $P(0.5 < X < 1.5)$

(b) Find $E(X)$ and $V(X)$

(c) Find mgf of the distribution

Solⁿ:

⑩ ① $P(X < 3) = ?$

we have,

$$P(X < 3) = \int_0^3 f(x) dx$$

$$= \int_0^3 \frac{9e^{-3x}}{\sqrt{2}} x^{2-1} dx$$

$$= 9 \int_0^3 e^{-3x} x^{2-1} dx$$

$$\int_0^1 e^{-ax} x^{n-1} dx = a^{-n} \Gamma(n)$$

$$= 9 \times \frac{\sqrt{2}}{3^2}$$

$$= \sqrt{2}$$

$$\boxed{P(X < 3) = 1}$$

①② $P(X > 3) = 1 - P(X < 3) = 1 - \sqrt{2} = 0$

we have,

$$P(X > 3) = \int_3^{\infty} f(x) dx$$

$$= \int_3^{\infty} \frac{9e^{-3x}}{\sqrt{2}} x^{2-1} dx$$

$$= 9 \int_3^{\infty} e^{-3x} x^{2-1} dx$$

$$\int =$$

Properties of Beta distribution of first kind:

$$(i) E(X) = \frac{m}{m+n}, \quad V(X) = \frac{mn}{(m+n)^2(m+n+1)}$$

$$(ii) \text{Skewness } (\beta_1) = \frac{4(m-n)^2(m+n+1)}{mn(m+n+2)^2}$$

$$(iii) \text{Kurtosis } (\beta_2) = \frac{3(m+n+1)[mn(m+n-6) + 2(m+n)^2]}{mn(m+n+2)(m+n+3)}$$

$$(iii) HM = \frac{m-1}{m+n-1} < AM$$

$$(iv) M_0 = \frac{m-1}{m+n-2} \quad \text{if } m > 1, n > 1$$

(v) If $m=1, n=1$, the beta distribution of first kind reduces to uniform distribution with $[0,1]$.

Beta distribution of 2nd kind:

A continuous random variable X is said to be followed a Beta distribution of second kind with parameters m and n if its pdf is defined as,

$$f(x) = \frac{1}{B(m, n)} \frac{x^{m-1}}{(1+x)^{m+n}}, \quad 0 < x < \infty, \quad m > 0, n > 0$$
$$= 0, \text{ otherwise.}$$

$$X \sim \beta_2(m, n)$$

Moments:

Let $X \sim \beta_2(m, n)$ then the r th moment about origin of X is given by;

$$\mu_r' = E(X^r) = \int_0^{\infty} x^r f(x) dx$$

$$= \int_0^{\infty} x^r \frac{1}{B(m, n)} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= \frac{1}{B(m, n)} \int_0^{\infty} \frac{x^{m+r-1}}{(1+x)^{m+n}} dx$$

$$= \frac{1}{B(m, n)} \int_0^{\infty} \frac{B(m+r, n) x^{m+r-1}}{(1+x)^{(m+r)+(n-r)}} dx$$

$$= \frac{1}{B(m, n)} B(m+r, n-r)$$

$$= \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \times \frac{\Gamma(m+r) \Gamma(n-r)}{\Gamma(m+n-r)}$$

$$= \frac{\Gamma(m) \Gamma(n)}{\Gamma(m) \Gamma(n)} \times \frac{\Gamma(m+r) \Gamma(n-r)}{\Gamma(m+n)}$$

$$\mu_1' = \frac{(m+1) \Gamma_{n-1}}{\Gamma_m \Gamma_n}$$

put $r=1$,

$$\mu_1' = \frac{(m+1) \Gamma_{n-1}}{\Gamma_m \Gamma_n} = \frac{m \Gamma_{n-1} \Gamma_{n-1}}{\Gamma_m (n-1) \Gamma_{n-1}}$$

$$\mu_1' = \frac{m}{n-1}$$

$$\therefore E(x) = \frac{m}{n-1}$$

$$\mu_2' = \frac{(m+2) \Gamma_{n-2}}{\Gamma_m \Gamma_n} = \frac{(m+1) m \Gamma_{n-2} \Gamma_{n-2}}{\Gamma_m (n-1)(n-2) \Gamma_{n-2}}$$

$$\mu_2' = \frac{m(m+1)}{(n-1)(n-2)}$$

Now,

$$V(x) = \mu_2' - (\mu_1')^2$$

$$= \frac{m(m+1)}{(n-1)(n-2)} - \left(\frac{m}{n-1}\right)^2$$

$$= \frac{m(m+1)}{(n-1)(n-2)} - \frac{m^2}{(n-1)^2}$$

$$= \frac{m}{(n-1)^2} \left[\frac{m+1}{n-2} - \frac{m}{n-1} \right]$$

$$= \frac{m}{n-1} \left[\frac{m+1 - m + m - 1 - m + 2m}{(n-1)(n-2)} \right]$$

$$V(x) = \frac{m(m+n-2)}{(n-1)^2(n-2)}$$

$$\frac{d}{dx} \frac{y}{\sqrt{u}} = \frac{y \frac{d}{dx} u - u \frac{d}{dx} y}{u^2}$$

Mode:

Let $x \in B_2(m, n)$ then its pdf is given as,

$$f(x) = \frac{1}{B(m, n)} \frac{x^{m-1}}{(1+x)^{m+n}}, \quad 0 < x < \infty$$

Mode is the solution of first $f'(x) = 0$ and $f''(x) < 0$.

now,

$$f'(x) = \frac{1}{B(m, n)} \frac{(1+x)^{m+n} (m-1) x^{m-2} - x^{m-1} (m+n) (1+x)^{m+n-1}}{(1+x)^{2(m+n)}}$$

$$= \frac{1}{B(m, n)} \frac{x^{m-2} (1+x)^{m+n-1} [(m-1)(1+x) - (m+n)x]}{(1+x)^{2(m+n)}}$$

$$f'(x) = \frac{1}{B(m, n)} \frac{x^{m-2} [m-1-(n+1)x]}{(1+x)^{m+n-1}}$$

Putting $f'(x) = 0$, we get,

$$m-1-(n+1)x = 0$$

$$\therefore x = \frac{m-1}{n+1} \quad \text{for } m > 1$$

It can also be shown that at $x = \frac{m-1}{n+1}$, $f''(x) < 0$

$$\therefore M_0 = \frac{m-1}{n+1} \quad \text{if } m > 1$$

$$M_0 = 0 \quad \text{if } 0 < m < 1.$$

HM:

Let $x \in B_2(m, n)$ then find HM.

$$\frac{1}{H} = E\left(\frac{1}{x}\right) = \int_0^\infty \frac{1}{x} f(x) dx$$

$$= \frac{1}{B(m, n)} \int_0^\infty \frac{1}{x} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= \frac{1}{\beta(m, n)} \int_0^1 \frac{x^{(m-1)-1}}{(1+x)^{m+n+1}} dx$$

$$= \frac{1}{\beta(m, n)} \beta(m-1, n+1)$$

$$= \frac{\Gamma(m+n)}{\Gamma(m)\Gamma(n)} \times \frac{\Gamma(m-1)\Gamma(n+1)}{\Gamma(m+n)}$$

$$= \frac{\Gamma(m-1)}{(m-1)\Gamma(m-1)} \cdot n\Gamma(n)$$

$$\frac{1}{H} = \frac{n}{m-1}$$

$$\therefore H = \frac{m-1}{n}$$

$$\therefore \text{Mean} > HM$$

Example: If x follows a distribution with pdf $f(x) = \frac{1}{4} x e^{-x/2}$, $0 < x < \infty$

$= 0$, otherwise

Obtain the mgf of the distribution and hence determine the mean & variance.